M9.3.b.1 Semantic Web and Ontologies

Tuto/Lab 3: SHACL validation

Part 1: Work environment

1- Download and install the last version of Protégé tool at
https://protege.stanford.edu
Ensure SHACL Editor Tab plugin is there.

Part 2: Running course samples

1- As a starting point, run the sample shapes and data graphs of the course material either using online SHACL playgroupd or local SHACL Tab in Protégé. For some examples, a link to playground session is already available on the slide. You may simply clic on it and run the validation. For other examples you may refer to Validating RDF Data book, chapter 5, available at http://book.validatingrdf.com/bookHtml011.html
2- For each example, run the validation and comment the obtained report. In case of validation error, correct the data graph and run validation again until you obtain a valid graph.

Part 3: Consistency validation of university ontology from the previous labs

In this part you should consider university.ttl ontology from the previous labs.

For the following sentences, write the SHACL shapes, run the validation, check the validation report, in case of error, correct the data graph, run again until you get a valid ontology. For some cases, you may need to add more instances to your data graph.

1- Ensure every **Course** has **at least one referenced ReadingMaterial**.
2- Ensure the property **hasReference** only links to individuals of type **ReadingMaterial**. Add an incorrect triple to check the vamlidation then remove or fix it.
3- From the previous lab we had : An **ExpertTeacher** is a Teacher with ≥1 expertise topic. Write SHACL shapes enforcing :
   a. Every **ExpertTeacher** must have **at least one hasExpertiseTopic** value.
   b. The value of **hasExpertiseTopic** must be a literal string OR an instance of Topic (depending on your ontology version).
   c. Ensure ExpertTeacher is always a subclass of Teacher using a SHACL condition shape (sh:condition). Here is an example

```
sh:condition ex:Rectangle ;
sh:condition [
        sh:property [
                sh:path ex:width ;
                sh:equals ex:height ;
        ] ;
] ;
```

4- Authors and Books
   a. Write a SHACL shape ensuring that every **Book** has **at least one author**.
   b. Ensure a **Book** has exactly one title.
   c. Ensure that the range of **authorOf** is a Book.
   d. Create a SPARQL-based SHACL constraint checking that if a Book has multiple authors they must be typed as CoAuthor.
5- Course Topics: From previous labs, a course has the same topics as its referenced Reading Materials. Now inforce the integrity using SHACL:
   a. Define a shape validating that every Course has a **hasTopic** property
   b. Write a constraint ensuring that a Course with topic “AI” uses a label containing “AI” (string matching).
   c. Create a SHACL rule copies topics from ReadingMaterial to Course.
6- Students and their Grades : From previous lab, A SuccessfulStudent is a student with grade ≥ 15. Add SHACL constraints to ensure the following :
   a. Grade values must be integers between 0 and 20.
   b. Every Student must have at least one grade.
   c. A SuccessfulStudent must have a grade ≥ 15.
7- Supervisor chains : From previous labs, A teacher A supervises any student working with student B who A already supervises. Add constraints to validate the following :
   a. Ensure any Student supervised by a Teacher must be typed Student
   b. Ensure there is no cycle in supervision relationships.
   c. A Teacher supervising ≥5 students must be typed SeniorSupervisor (create class).